

Electricity Consumption Forecasting

Decision Tree and XGBoost for hourly load prediction

MEE344 Machine Learning – Term Project

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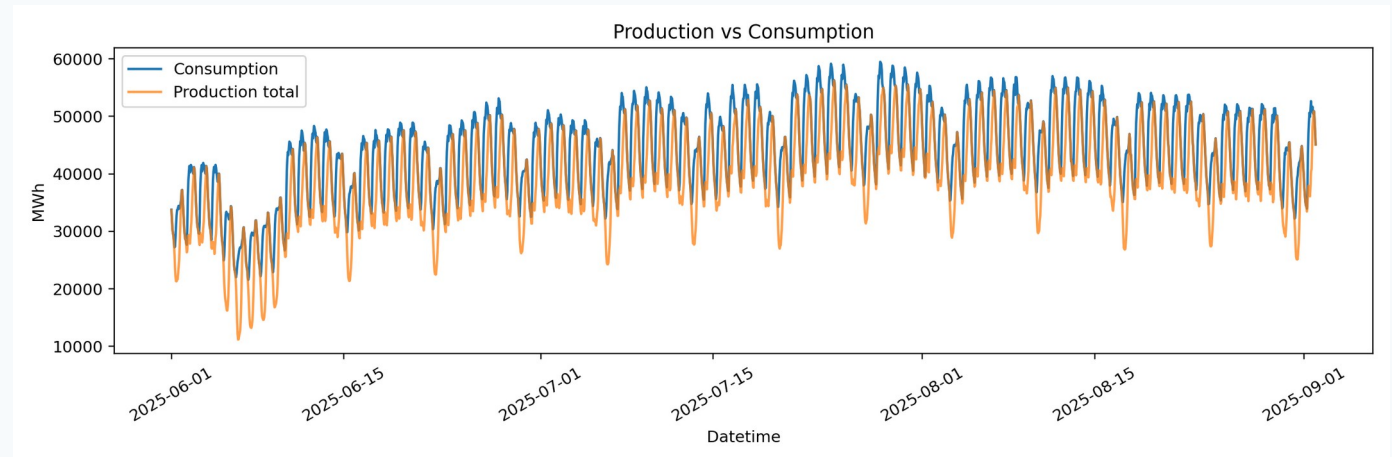
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Dataset window: 1 Jun 2025 - 1 Sep 2025

Target: hourly electricity consumption (Tuketim)



Problem Definition

A supervised regression task with explicit forecasting constraints



We predict hourly electricity consumption. The target is continuous, so the task is regression.

Same-hour production is excluded from the main model because it would not be known before forecasting that hour.

The model uses past consumption, past production and known calendar information.

Dataset Overview

Two hourly CSV files are merged by timestamp

2232

raw production rows

2232

raw consumption rows

2064

modeled samples

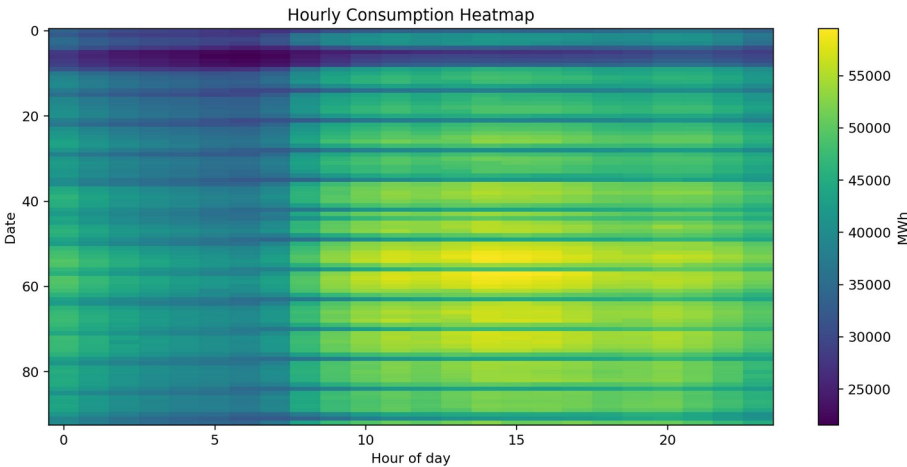
309

test samples

37

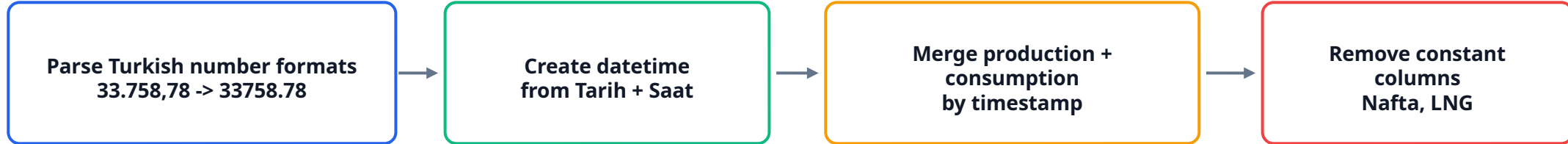
features used

Item	Status
Time range	2025-06-01 00:00 to 2025-09-01 23:00
Frequency	Hourly, no gaps detected
Merge key	datetime
Target	Tuketim (MWh)
Constant columns removed	Nafta, LNG



Preprocessing Decisions

Data cleaning was kept explicit and reproducible



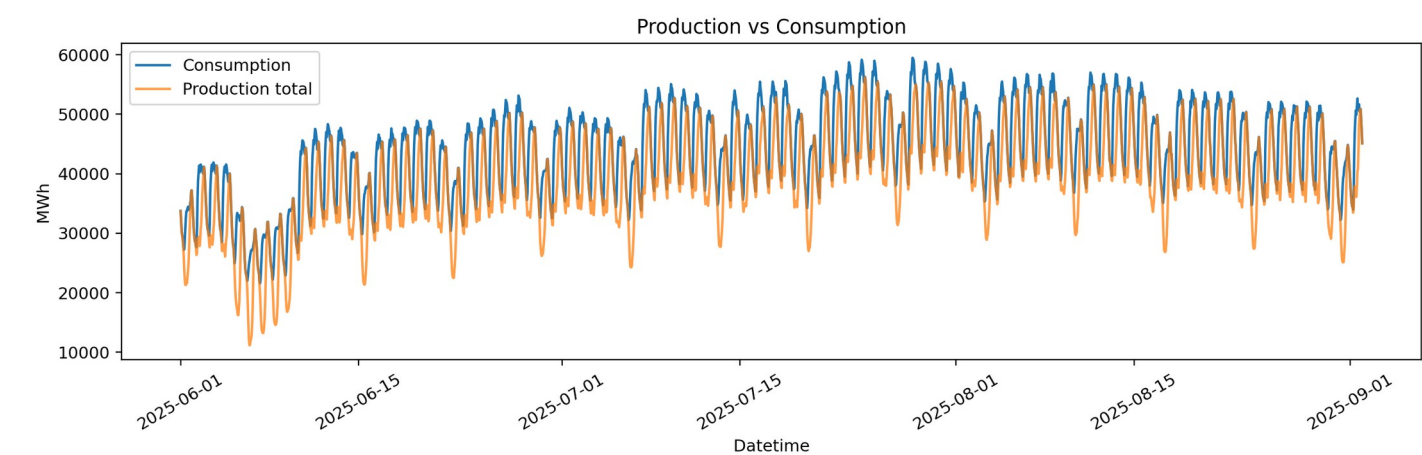
No raw missing values were found. Missing rows introduced by lag/rolling features are dropped after feature construction.

No nominal categorical variables were present. Date/time strings are encoded as numeric calendar and cyclic features.

No standardization is applied: Decision Tree and XGBoost are tree-based methods and do not require feature scaling.

Exploratory Analysis: System Load

Consumption and production are correlated but not identical

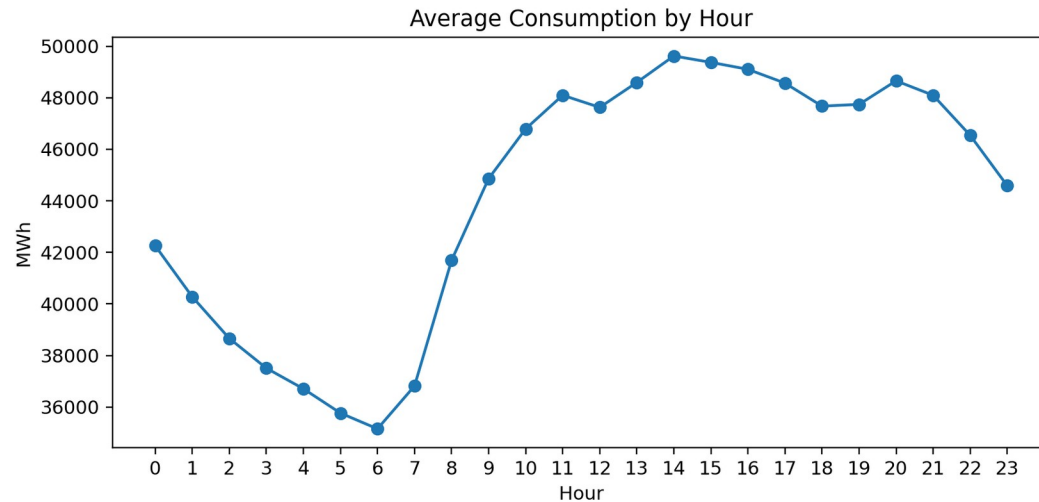


Observation	Implication
Production total is not equal to consumption	Do not treat Toplam as target substitute
Consumption follows daily demand cycles	Use calendar features
Lag autocorrelation is strong	Use lag and rolling features

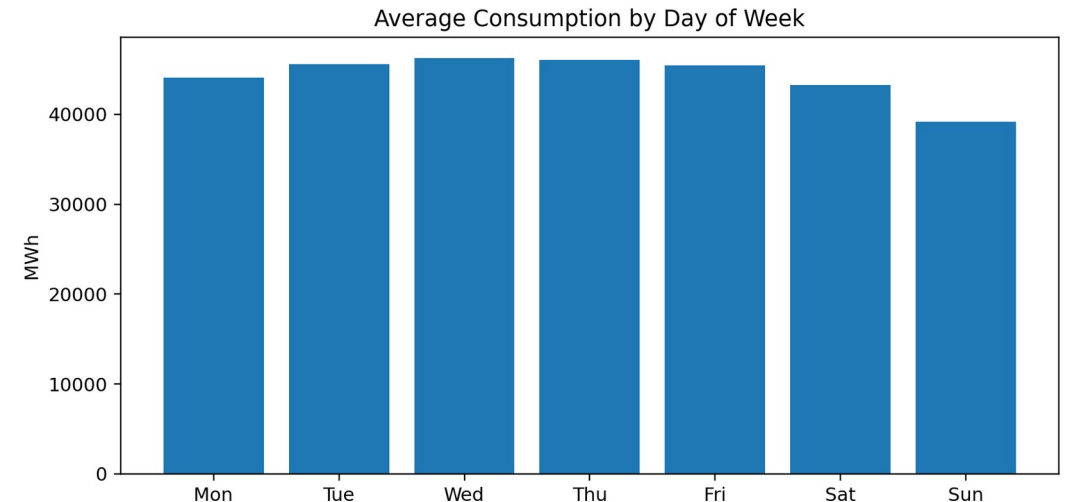
Key modeling choice: same-time production values are excluded from the main forecast to avoid feature availability leakage.

Exploratory Analysis: Temporal Patterns

Hourly and weekly structure justify calendar features



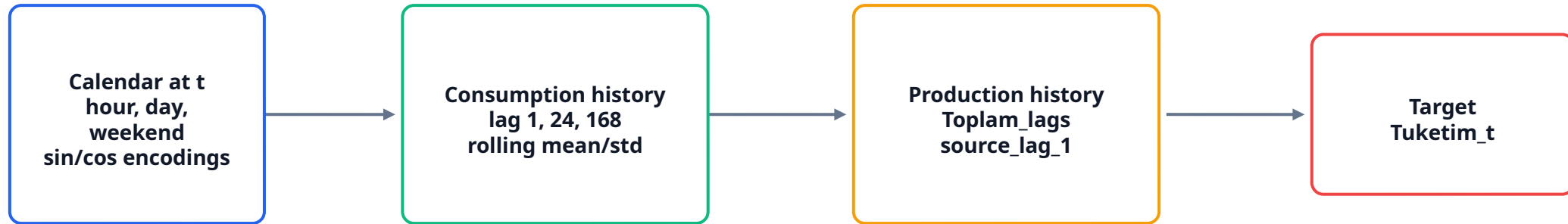
Average consumption rises after early morning and remains high during daytime/evening.



Weekends, especially Sunday, show lower average consumption than weekdays.

Feature Engineering

Rows represent prediction time t ; features use $t-1$ or earlier

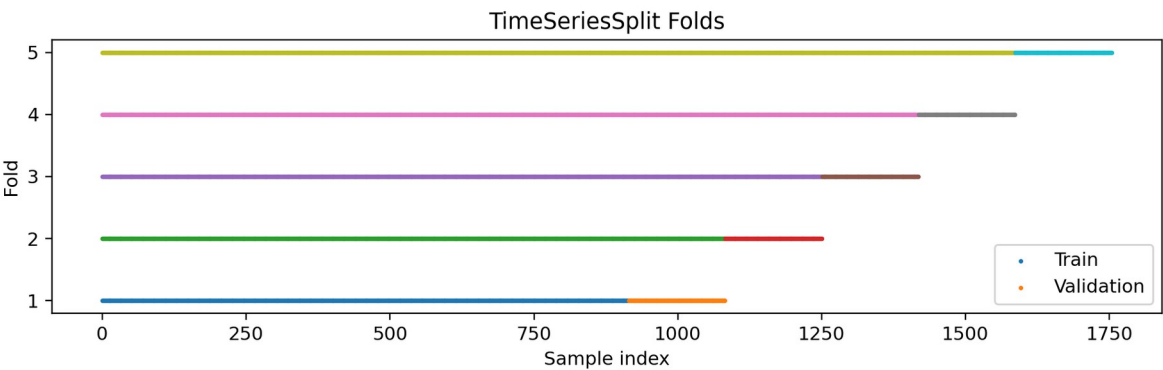


Lag 1: previous hour. Lag 24: same hour yesterday. Lag 168: same hour last week.

Rolling features are computed as `shift(1).rolling(...)`, so the target row is never included.

Validation Methodology

The test period is untouched until final evaluation

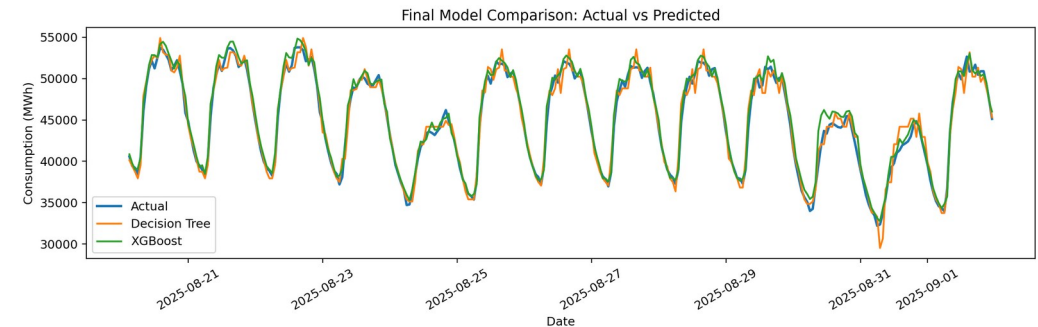
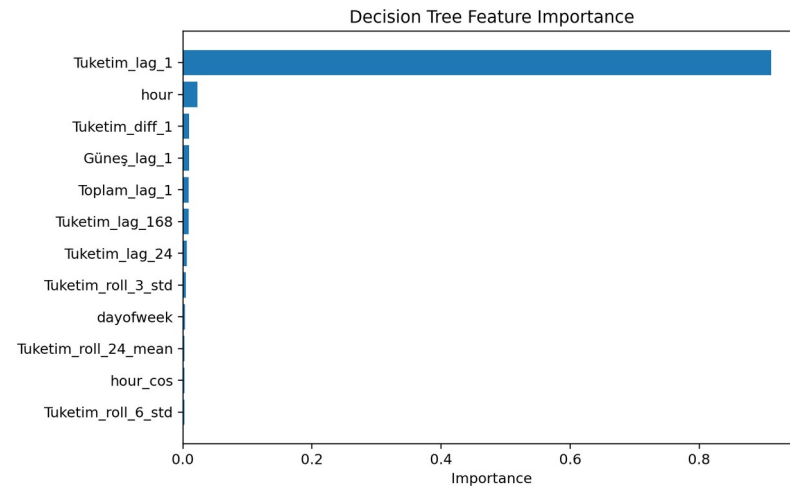


Split	Purpose
Earlier 85%	train + validation region
Last 15%	final test set
TimeSeriesSplit	hyperparameter tuning
Metrics	RMSE, MAE, R2

Random split is avoided because it can train on future samples and validate on past samples. TimeSeriesSplit preserves chronological order.

Model 1: Decision Tree Regressor

Interpretable baseline with controlled complexity



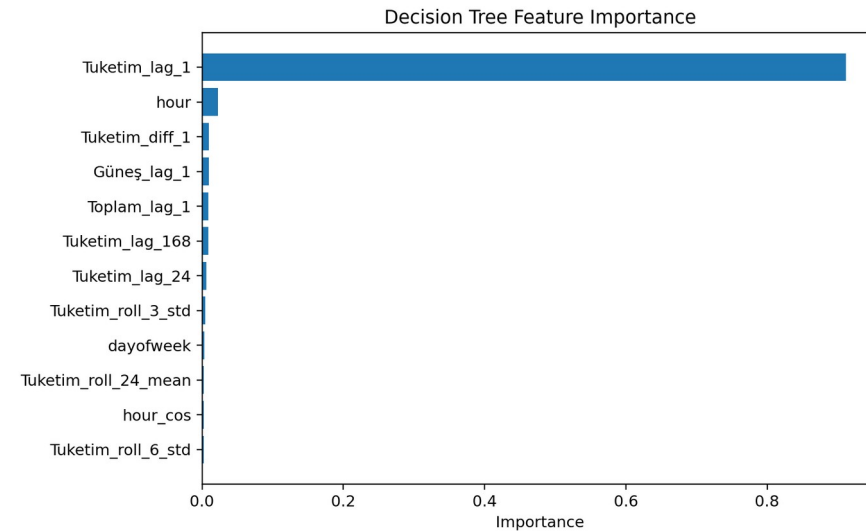
Decision Tree Results

Tuning reduces complexity but only slightly improves RMSE

Model	RMSE	MAE	R2
Baseline	1011.92	739.69	0.9688
Tuned	1004.10	759.15	0.9693

Best tuned configuration: max_depth=10,
min_samples_leaf=5, min_samples_split=2.

The model is interpretable, but it is still a single
tree; boosted trees are expected to generalize
better.

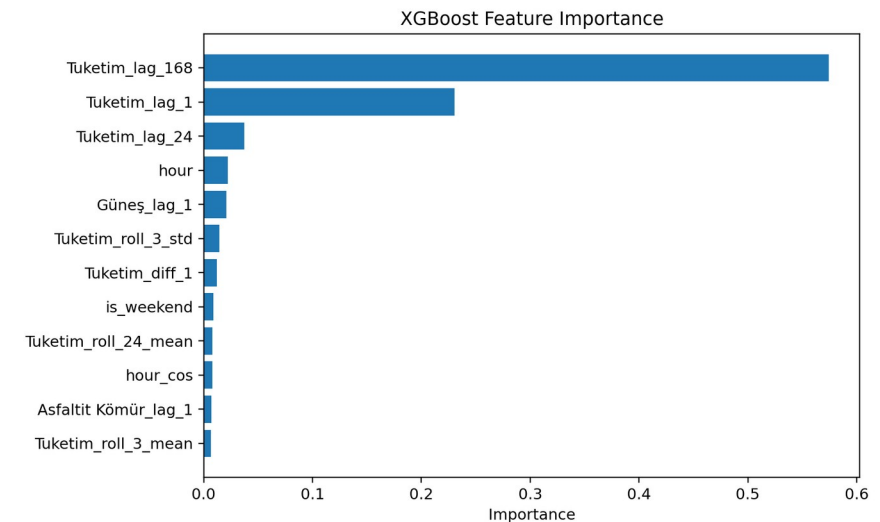


Model 2: XGBoost Regressor

Boosted tree ensemble for stronger nonlinear prediction



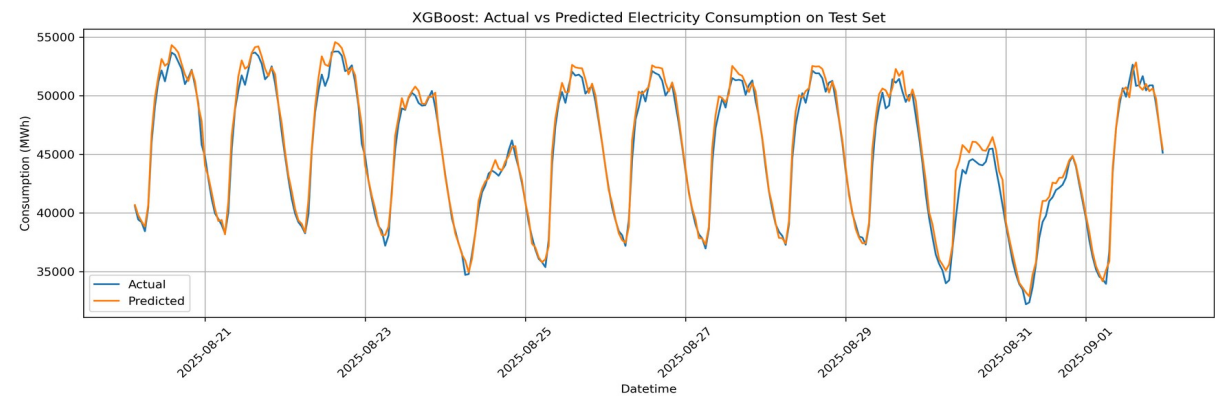
Final tuned parameters: 200 trees, depth 3, learning rate 0.03, subsample 0.9, colsample 0.9.



XGBoost Results

XGBoost achieves the lowest final test error

Model	RMSE	MAE	R2
Baseline	852.22	647.50	0.9779
Tuned	784.41	591.91	0.981

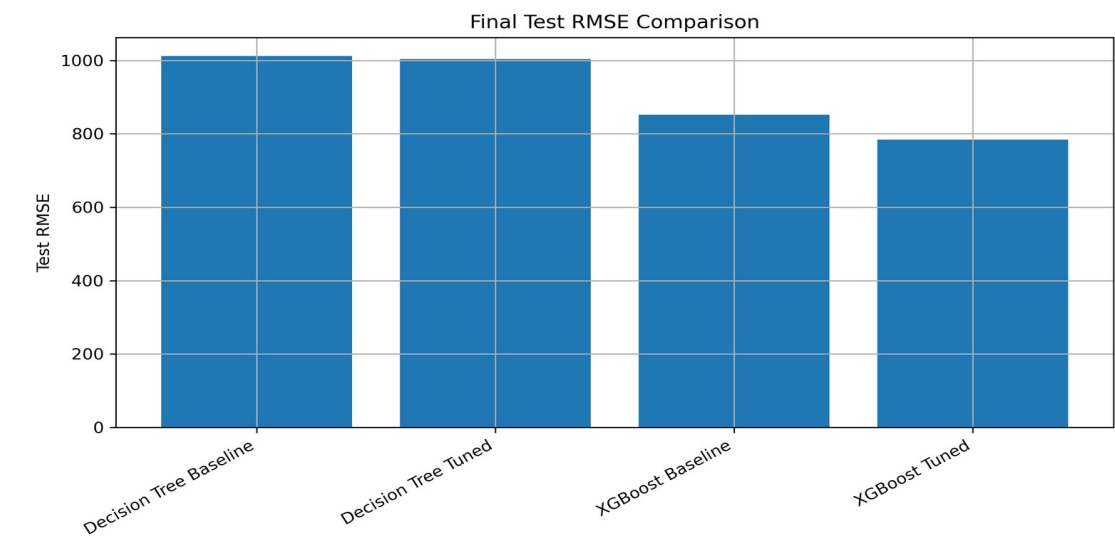


The tuned XGBoost model improves over the tuned Decision Tree by about 21.9% RMSE.

The remaining errors likely correspond to rapid local changes not captured by weather or holiday variables.

Final Model Comparison

Both models perform well; XGBoost is strongest



Model	RMSE	MAE	R2
Decision Tree Tuned	1004.10	759.15	0.9693
XGBoost Tuned	784.41	591.91	0.981

21.9%

RMSE reduction

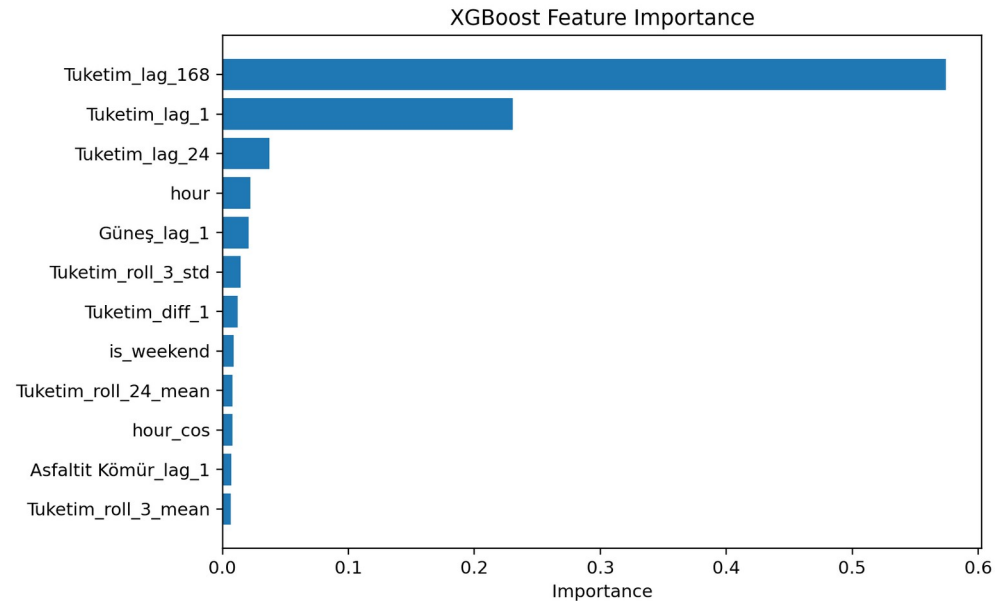
0.981

Best R2

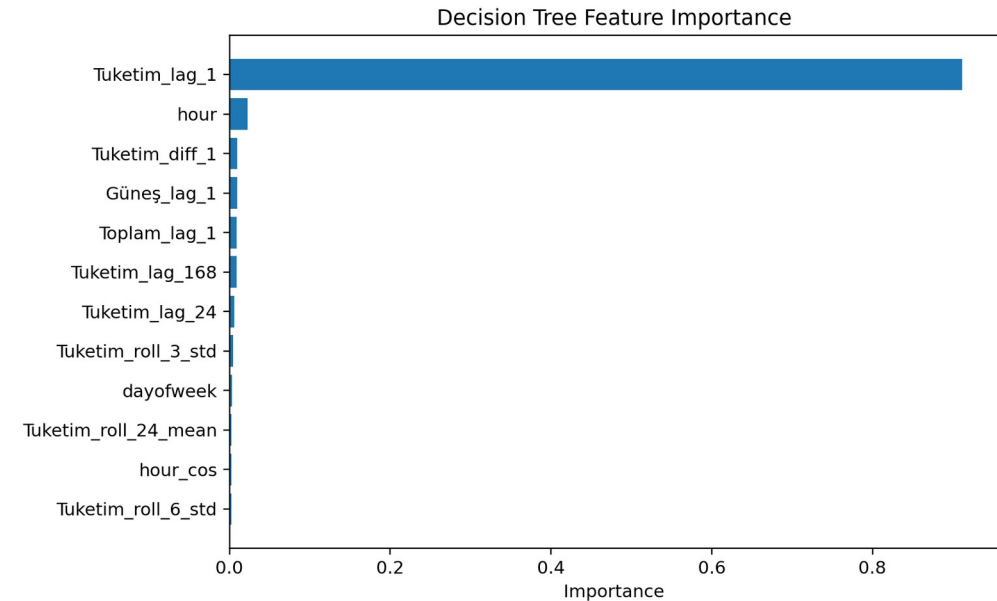
Final conclusion: XGBoost is selected as the better model under the same leakage-safe feature set and test period.

Insights and Interpretation

The strongest signals are recent consumption and system state



Lagged consumption features confirm strong short-term and daily dependence in electricity demand.



Lagged production features provide recent system-state information, but are kept lagged to avoid availability leakage.

Conclusion and Discussion

What the model results imply and where the setup is limited

Final conclusion

XGBoost is selected as the best model under the same leakage-safe feature set and final test period.

RMSE: 784.41 MAE: 591.91 R^2 : 0.981

Compared with the tuned Decision Tree, XGBoost reduces RMSE by about 21.9%.

Interpretation

The dominant features are lagged consumption values, especially the weekly lag and previous-hour load.

This matches electricity demand behavior: recent load and weekly routines carry strong predictive information.

Lagged production contributes as recent system-state information, not as same-hour leakage.

Limitations

The dataset covers only the summer period from June to September 2025. Weather, public holidays, electricity prices and industrial activity are not included. Sudden demand changes remain harder to predict with only lagged historical features.

Main takeaway: a leakage-safe time-series setup with lagged features gives a realistic forecasting task; boosted trees improve accuracy while preserving interpretable feature importance.